

Description and choice data for the domain “Colour combos”

Description of the choice domain 8, Colour combos

The prompt question and the universe of five response options in the choice domain **Colour combos** are as follows. The labels a , b , c , d and e were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

The source for this domain is the website “The top tens”, page “Two colors that look good side by side.” The color combinations here are ranked 1, 4, 5, 13 and 14. We chose a selection of five high ranking combinations among which there were many colors in common. Using a similarity measure equal to the number of colours in common between two pairs, there are two doubleton choice sets where the two colour pairs have no colours in common ($\{a, e\}$ and $\{b, d\}$) and eight where the two colour pairs have one colour in common. This gives six tripleton pairs in which one might expect a similarity effect. The name for this domain used in data files and documents, “Colour combos” is an abbreviation of the title seen by participants, “Colour combinations”.

Which one of these colour combinations do you like best?

- Black and red
- Black and purple
- Black and blue
- Blue and red
- Blue and purple

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

Colour combinations

Which one of these colour combinations do you like best?

☐ Blue and purple

☐ Black and red

☐ Black and purple

☐ Blue and red

☐ Black and blue

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Figure 1: Screenshot for domain Colour combos

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	18	22	-	-	-	0.450	0.550	-	-	-
$\{a, c\}$	21	-	19	-	-	0.525	-	0.475	-	-
$\{b, c\}$	-	19	21	-	-	-	0.475	0.525	-	-
$\{a, b, c\}$	18	7	15	-	-	0.450	0.175	0.375	-	-
$\{a, d\}$	25	-	-	15	-	0.625	-	-	0.375	-
$\{b, d\}$	-	19	-	21	-	-	0.475	-	0.525	-
$\{a, b, d\}$	22	7	-	11	-	0.550	0.175	-	0.275	-
$\{c, d\}$	-	-	21	19	-	-	-	0.525	0.475	-
$\{a, c, d\}$	20	-	11	9	-	0.500	-	0.275	0.225	-
$\{b, c, d\}$	-	6	17	17	-	-	0.150	0.425	0.425	-
$\{a, b, c, d\}$	13	7	11	9	-	0.325	0.175	0.275	0.225	-
$\{a, e\}$	21	-	-	-	19	0.525	-	-	-	0.475
$\{b, e\}$	-	19	-	-	21	-	0.475	-	-	0.525
$\{a, b, e\}$	12	12	-	-	16	0.300	0.300	-	-	0.400
$\{c, e\}$	-	-	19	-	21	-	-	0.475	-	0.525
$\{a, c, e\}$	15	-	15	-	10	0.375	-	0.375	-	0.250
$\{b, c, e\}$	-	14	14	-	12	-	0.350	0.350	-	0.300
$\{a, b, c, e\}$	12	5	13	-	11	0.293	0.122	0.317	-	0.268
$\{d, e\}$	-	-	-	21	19	-	-	-	0.525	0.475
$\{a, d, e\}$	19	-	-	13	8	0.475	-	-	0.325	0.200
$\{b, d, e\}$	-	17	-	15	8	-	0.425	-	0.375	0.200
$\{a, b, d, e\}$	10	8	-	16	7	0.244	0.195	-	0.390	0.171
$\{c, d, e\}$	-	-	14	11	15	-	-	0.350	0.275	0.375
$\{a, c, d, e\}$	14	-	5	10	11	0.350	-	0.125	0.250	0.275
$\{b, c, d, e\}$	-	9	14	7	10	-	0.225	0.350	0.175	0.250
$\{a, b, c, d, e\}$	13	7	2	6	12	0.325	0.175	0.050	0.150	0.300

Table 1: Observed choice counts and proportions.

Choice data for domain 8, Colour combos

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

